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Two Genetic Loci associated with Medial Collateral Ligament Injury

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ABSTRACT

Medial collateral ligament (MCL) injuries are a common knee injury, especially in competitive athletes. Identifying genetic loci associated with MCL injury could shed light on its etiology. A genome-wide association screen was performed using data from the Research Program in Genes, Environment and Health (RPGEH) including 1 572 cases of MCL injury and 100 931 controls. 2 SNPs (rs80351309 and rs6083471) showed an association with MCL injury at genome-wide significance ($p < 5 \times 10^{-8}$) with moderate effects (odds ratios = 2.12 and 1.57, respectively). For rs80351309, the genotypes were imputed with only moderate accuracy, so this SNP should be viewed with caution until its association with MCL injury can be validated. The SNPs rs80351309 and rs6083471 show a statistically significant association with MCL injury. It will be important to replicate this finding in future studies.

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Introduction

The medial collateral ligament (MCL) of the knee functions as a valgus stabilizer. Injuries of the MCL are a common knee injury; in a study of students at a United States Military Academy, the incidence rate of isolated MCL injury of any grade was 7.27 per 1 000 person years [25]. MCL sprains are typically caused by valgus or tibial external rotation forces delivered to the knee. MCL sprains occurred with highest frequency among athletes competing in wrestling, hockey, judo and rugby [25].

Although environmental factors such as participation in cutting sports play a dominant role in MCL sprains, an unanswered question is whether there may also be genetic differences amongst athletes that affect their inherent risk for MCL sprain. To date, there have been no studies on genetic factors for MCL injuries.

To begin to address this question, we performed a genetic association study for MCL injury. We identified individuals who had suffered an MCL sprain or rupture from a cohort of 102 503 patients in the Research Program on Genes, Environment, and Health (RPGEH) of the Kaiser Permanente Health Plan of Northern California (KPNC). A gene association analysis revealed 2 loci associated with MCL injury with genome-wide significance.

Methods

A genome-wide association screen was performed for MCL injury using data from the genotyped Genetic Epidemiology Research on Adult Health and Aging (GERA) cohort of the RPGEH. The data generation and data analysis pipeline have been previously described

in Roos et al. 2016 [26]. A complete description of the cohort and study design can be found in dbGaP (Study Accession: phs000674.v1.p1).

Our analysis cohort (n = 102 503) included 59 479 females, 42 958 males, and 66 individuals of uncertain sex. Moreover, our analysis cohort was ethnically diverse, including 83 264 European-White (EUR); 8 560 Latino (LAT); 7 518 East Asian (EAS) and 3 161 African-American (AFR) individuals based on ancestry principle components.

Participants were genotyped at over 650 000 SNPs on 4 race/ethnicity-specific Affymetrix Axiom genome-wide arrays optimized for individuals of European (EUR), African-American (AFR), East Asian (EAS), and Latino (LAT) race/ethnicity [9, 10]. The final number of SNPs that were directly genotyped was 670 572 for EUR; 802 186 for LAT; 708 373 for EAS; and 878 176 for AFR arrays.

Genotypes were pre-phased with Shape-IT v2.r644 (https://mathgen.stats.ox.ac.uk/genetics_software/shapeit/shapeit.html; accessed Feb. 2, 2016) then imputed to a cosmopolitan reference panel consisting of all individuals from the 1000 Genomes Project (Mar 2012 release) using IMPUTE2 v2.2.2 (https://mathgen.stats.ox.ac.uk/impute/impute_v2.html; accessed Feb. 2, 106) and standard procedures with a cutoff of $R^2 > 0.3$ [11–13]. The quality of the imputed data was previously validated in Jorgenson et al., 2015 [15]. The final number of SNPs that were imputed was 9 207 988 for EUR (9 878 560 total); 10 380 912 for LAT (11 183 098 total); 8 355 578 for EAS (9 063 951 total); and 16 659 640 for AFR arrays (17 537 816 total).

Determination of genetic ancestry was performed by principal component analysis (PCA), as previously described [26]. These ancestry principal components were used in the GWAS to adjust for genetic ancestry.

Phenotype definition

MCL injury cases were identified in the GERA cohort based on clinical diagnoses and surgical procedures captured in the KPNC electronic health record system. The electronic health record includes reported injuries over the entire lifetime of the patients, including those that occurred prior to enrollment in KPNC as well as those that occurred after the genotyping analysis was performed, if reported by the patient and recorded by the physician.

Cases were defined as individuals with at least one International Classification of Disease, Ninth Revision (ICD-9) code: 844.1 (sprain of medial collateral ligament of knee; 1 555 cases) or 717.82 (old disruption of medial collateral ligament; 22 cases). The main difference between 844.1 and 717.82 is that 717.82 describes a pre-existing or chronic injury whereas 844.1 indicates a current injury. 5 cases had both ICD717.82 and CPT844.1 codes, bringing the total number of cases to 1 572. The ICD-9 codes were assigned by the physician for each patient in the KPNC system. MCL injuries are graded into 3 groups based on MRI: grade 1: (minor sprain) high signal is seen medial (superficial) to the ligament, which looks normal, grade 2: (severe sprain or partial tear) high signal is seen medial to the ligament, with high signal or partial disruption of the ligament and grade 3: complete disruption of the ligament. Both of the ICD-9 codes include all grades of MCL injury, hence we cannot distinguish sprains (grades 1 and 2) from full rupture (grade 3) in either ICD-9 code.

Genome-wide association and meta-analysis

Genome-wide association analyses of the GERA cohort genomic data were conducted using PLINK v1.90(b3.34) (https://www.cog-genomics.org/plink2; accessed Feb. 1, 2016) [2, 23]. SNP associations were tested with MCL injury with a logistic regression model using allele counts for typed and imputed SNPs in an additive genetic model for each of the race/ethnic populations. The model was adjusted for genetic sex, age at enrollment into the RPGEH cohort, race/ethnicity using principal components, and variations in genotyping protocol. The variations in genotyping protocol include: chip type for all populations (refers to the Affymetrix chip version), genotyping package (refers to the set of chips that were processed together) and reagent kit (refers to A vs O reagent kit distributed by Affymetrix) when there was variation in the individuals of a specific population. The final number of SNPs that were analyzed was 8 795 348 for EUR; 9 153 118 for LAT; 8 055 053 for EAS and 14 989 676 for AFR populations. To account for inflation due to population stratification, the genomic control parameter (λgc) was calculated: EUR (.994), LAT (1.01), EAS (.975), AFR (.891). λgc is defined as the median of the resulting chi-squared test statistics divided by the expected median of the chi-squared distribution [6]. Subsequently, p-values were adjusted for genomic control in each population. Results from each population were combined by inverse-variance, fixed-effects meta-analysis as previously described [26]. The final number of SNPs analyzed in the fixed-effects meta-analysis was 8 668 382. Power calculations were made using the software at http://csg.sph.umich.edu/abecasis/cats/gas_power_calculator/index.html; accessed Dec. 20, 2016 [28].

We examined the level of heterogeneity using 2 measures: 1) the I^2 statistic, which measures the percentage of variability across ancestry groups that is due to heterogeneity, where a lower value indicates more consistent results across races, and 2) Cochran's Q statistic, which measures whether observed differences in results between different races are due to chance alone, where a low associated p-value indicates heterogeneity [3, 8]. The 95 % confidence interval for I^2 was calculated using the heterogi module for STATA.

Further bioinformatics investigation of the top genome-wide significant loci from the meta-analysis was conducted. Regional association plots were generated for each locus with LocusZoom (http://locuszoom.sph.umich.edu/locuszoom/; accessed July 1, 2016) [22]. The genomic context of each SNP was investigated using RegulomeDB (http://regulomedb.org/; accessed June 1, 2016)[1] web tools. Whether each SNP is an expression quantitative trait locus (eQTL) was queried using the NCBI eQTL Browser (http://www.ncbi.nlm.nih.gov/projects/gap/eqtl/index.cgi; accessed June 2, 2016) and the Genotype-Tissue Expression (GTEx) Portal (http://www.gtexportal.org/home/; accessed June 4, 2016). ChIP seq data from the ENCODE project was used to determine whether SNPs were located within transcription factor binding sites [4].

Summary statistics for all SNPs from the fixed-effects meta-analysis are available at https://grasp.nhlbi.nih.gov/FullResults.aspx.

Ethical considerations

This study analyzed stored data from RPGEH subjects who consented to genomic testing and use of their genomic data, as well as health data from the KPNC electronic health record, for future re-

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▶ **Table 1** Demographic factors of the GERA study population used in genome-wide association analyses of MCL injury.

	Cases[a]	Controls	Overall
Subjects (%)	1 572 (1.5 %)	100 931 (98.5 %)	102 503
Sex (%)[b]			
Female	922 (1.6 %)	58 557 (98.4 %)	59 479
Male	649 (1.5 %)	42 309 (98.5 %)	42 958
Undetermined	1 (1.5 %)	65 (98.5 %)	66
Race (%)[c]			
European	1 280 (1.5 %)	81 984 (98.5 %)	83 264
Latin American	153 (1.8 %)	8 407 (98.2 %)	8 560
East Asian	85 (1.1 %)	7 433 (98.9 %)	7 518
African American	54 (1.7 %)	3 107 (98.3 %)	3 161
Age[d]	60.9 (60.3-61.5)	62.7 (62.6–62.8)	62.6 (62.5–62.7)

[a] Cases with MCL injury as defined by individuals with 1 or more qualifying ICD-9 or CPT-4 code in their EHR. For details, see Methods

[b] Sex/gender as determined by an individual's genetic data, reported as the number and percentage of total cases, controls, or overall for each respective group. For details see Methods and dbGaP (Study Accession: phs000674.v1.p1)

[c] Race/ethnic groups as determined by PCA on an individual's genetic data from the GERA cohort. Reported as the number and percentage of total cases, controls, or overall for each respective group

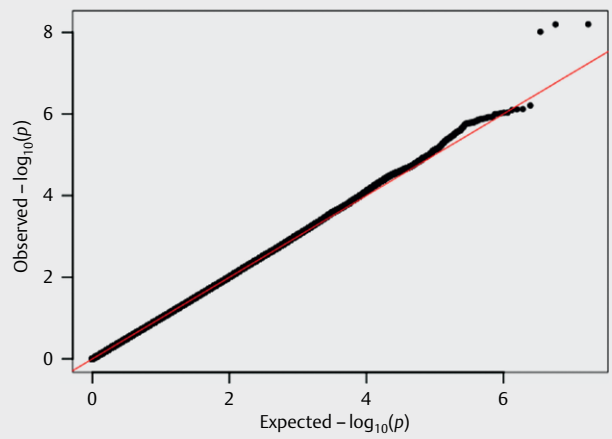
[d] Age at subject enrollment in the GERA cohort, reported as mean age with 95 % CI. The difference in ages between cases and controls is highly significant ($p = 1.5 \times 10^{-8}$)

search studies. The health and genotype data for the subjects were de-identified. All study procedures were approved by the Kaiser Foundation Research Institute Institutional Review Board. This study meets the ethical standards of the journal [7].

Results

Study population and genotype information

We performed a logistic regression for SNPs associated with MCL injury using genotype and medical data from 102 503 individuals that included 1 572 cases and 100 931 controls (▶ **Table 1**). Overall, the period prevalence of MCL injury was 1.5 %. Participation in sports was not included in the electronic health records, and hence we were not able to determine the incidence rate for the subset of the population that are athletes. There was a small difference in the age of enrollment between the cases and controls (1.8 years) that was statistically significant ($p = 1.5 \times 10^{-8}$) (▶ **Table 1**). One possibility is that this might be caused by an ascertainment bias where some elderly patients that enrolled in the RPGEH program might be systematically missed as cases if they incurred the MCL injury when they were young, before electronic records were in common practice.



▶ **Fig. 1** Quantile-quantile plot for genome-wide association analyses of MCL injury. The expected vs. observed log transformed values for the 8 668 382 p-values from the meta-analysis are graphed. The y-axis shows the observed p-values and the x-axis shows the p-values expected by chance. The black dots represent the SNPs arranged by their observed p-values and the red line shows the expected trajectory if the SNPs had p-values expected by chance.

Genome-wide study for association with MCL injury

We compared the observed p-values to the distribution of p-values expected by chance in a Q-Q plot (▶ **Fig. 1**). The black dots deviate from the red line for the 3 lowest observed p-values in the upper right hand corner, suggesting that the observed association signals for these most extreme SNPs are significantly different than the signals that would be expected by chance.

The p-value for every SNP from the meta-analysis is shown in a Manhattan plot in ▶ **Fig. 2**. Using $p = 5 \times 10^{-8}$ as a cut-off for genome-wide significance, 2 independent loci showed a genome-wide significant association with MCL injury. The first locus contains rs80351309 on chromosome 4 and the second contains rs6083471 and rs79027201 on chromosome 20. rs6083471 and rs79027201 are located about 15 kb from each other on chromosome 20, with $R^2 = 0.99$ (▶ **Table 2**). Hence, it is likely that these 2 SNPs represent one association signal from a linkage disequilibrium block on chromosome 20, and we refer to this locus using rs6083471 hereafter.

Neither rs80351309 nor rs6083471 were directly genotyped on the Affymetrix chips, but rather their genotype data was imputed (Supplemental **Table S1**). For rs6083471, the overall R^2 value was 0.97, indicating that the genotype was fairly accurate using imputation. For rs80351309, the overall R^2 value was 0.56 indicating

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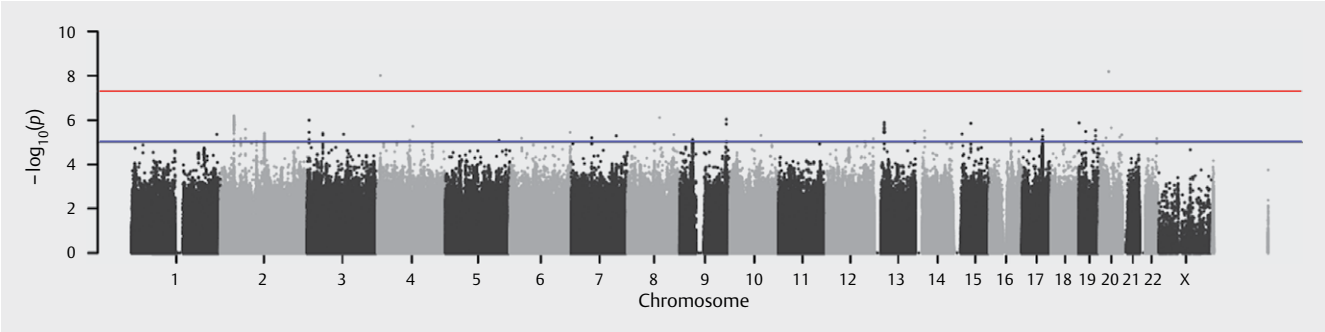

▶ **Fig. 2** Manhattan plot for genome-wide association analyses of MCL injury. The -$\log_{10}$ p-values for association with MCL injury for SNPs from the meta-analysis are plotted by genomic position with chromosome number listed across the bottom. The y-axis shows the -$\log_{10}$ p-value for association with MCL injury. The blue line represents suggestive genome-wide significance ($p < 5 \times 10^{-5}$) and the red line represents genome-wide significance ($p < 5 \times 10^{-8}$).

▶ **Table 2** Genome-wide association analyses for MCL injury.

SNP	Gene	EA[a]	EAF[b]	P-value[c]	OR (95 % CI)[d]
rs80351309[e]	GRPEL1 FLJ36777	T	0.012	6.8×10^{-9}	2.12 (1.86-2.38)
rs6083471[e]	FLJ33581	C	0.043	6.3×10^{-9}	1.57 (1.42-1.72)
rs79027201[e,f]	FLJ33581	C	0.040	6.4×10^{-9}	1.58 (1.43-1.73)

[a] Effect allele (minor allele)

[b] Effect allele frequency

[c] P-value from fixed-effects meta-analysis

[d] Allelic odds ratio with 95 % confidence interval

[e] Genotypes for these SNPs were imputed: rs80351309 ($R^2 = 0.56$), rs6083471 ($R^2 = 0.97$) and rs79027201 ($R^2 = 0.99$)

[f] rs79027201 is tightly linked to rs6083471 ($R^2 = 0.99$, D' = 1.00)

▶ **Table 3** Genotype distributions for rs80351309 and rs6083471.

rs80351309	**T/T**	**T/C**	**C/C**
Cases	0	62	1 370
Controls	12	1 947	90 061
Overall	12	2 009	91 431
Risk for MCL injury	0.0000	0.0309	0.0150
Relative risk for MCL injury[a]	NA	2.06	1.00
rs6083471	**C/C**	**C/T**	**T/T**
Cases	4	175	1 385
Controls	144	7 281	93 119
Overall	148	7 456	94 504
Risk for MCL injury	0.0270	0.0235	0.0147
Relative risk for MCL injury[a]	1.84	1.60	1.00

[a] Risk relative to individuals homozygous for the protective allele

that the imputed genotype differs substantially from the true genotype. The imputation quality for rs80351309 ranged from $R^2 = 0.55$ in the EUR ancestry group to $R^2 = 0.63$ in the EAS ancestry group. Even though the imputed genotype for rs80351309 may be somewhat inaccurate, it still showed an association with MCL injury that was genome-wide significant.

For rs80351309, the frequency of the risk allele (T) ranged between 1.1 % in the European controls to 1.5 % in the African controls. The overall allelic odds ratio was 2.12, indicating that this moderately rare SNP also has a moderate effect size (▶ **Table 2**). Individuals that carried one copy of the risk allele for rs80351309 (T/C) had a 2.06-fold increased risk of MCL injury compared to individuals with no risk alleles (C/C) (▶ **Table 3**).

For rs6083471, the frequency of the risk allele (C) was 4.3 % in European controls and 3.1 % in Latin-American controls. The overall allelic odds ratio was 1.57 (▶ **Table 2**). Individuals carrying one copy of the risk allele (C/T) had a 1.60-fold higher risk for MCL injury compared to individuals with no risk alleles (T/T) (▶ **Table 3**). Individuals with 2 risk alleles (C/C) had a 1.84-fold increased risk for MCL injury compared to individuals with 2 protective alleles (T/T) (▶ **Table 3**).

The GWAS results were analyzed to determine whether the association with MCL injury for either rs80351309 or rs6083471 was stronger in some races than in others, a phenomenon known as heterogeneity [14]. ▶ **Table 4** shows the p-values and odds ratios for these 2 SNPs for each of the races. For both SNPs, the smallest p-value was observed for the European population, which is expected as 81 % of the cohort was European. The odds ratios for each race were in the same direction and of similar magnitude. Using I^2 and Cochran's Q to assess heterogeneity, we saw no evidence of heterogeneity for either SNP (▶ **Table 4**). The logistic regression analysis did not converge on a p-value for one race (EAS) for rs80351309 and for 2 races (AFR and EAS) for rs6083471.

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▶ **Table 4** Association statistics for rs80351309 and rs6083471 in individual races.

Race	SNP	EA[a]	P-value[b]	OR (95 % CI)[c]	I^2 (95 % CI)[d]	Q [e]
EUR	rs80351309	T	5.63×10^{-7}	2.05 (1.55 - 2.71)	0.00 (0.0 -.90)	0.82
LAT	rs80351309	T	3.55×10^{-2}	2.32 (1.06 - 5.05)		
AFR	rs80351309	T	3.61×10^{-2}	2.83 (1.01 - 7.91)		
EUR	rs6083471	C	1.44×10^{-8}	1.58 (1.35 -1.86)	0.00	0.82
LAT	rs6083471	C	1.74×10^{-1}	1.49 (.84 - 2.63)		

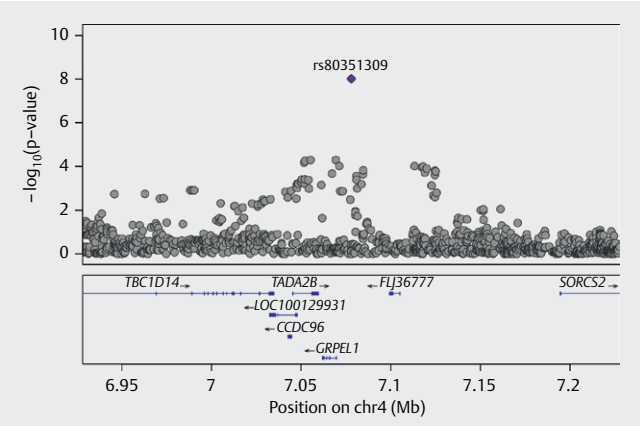
[a] Effect allele

[b] P-value adjusted for lambda genomic inflation factor from logistic regression

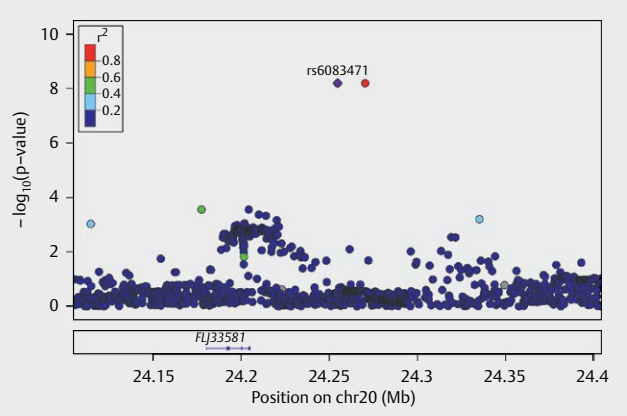
[c] Allelic odds ratio (95 % confidence interval)

[d] I^2 - percentage of variability between races that is due to heterogeneity (95 % confidence interval)

[e] Cochran's Q, p-value that the association is different between races



▶ **Fig. 3** Regional-association plot for rs80351309 with MCL injury. Tested SNPs are arranged by genomic position on chromosome 4 (x-axis) in a 300 kb window around the lead SNP rs80351309 (purple diamond). The y-axis indicates -$\log_{10}$ p-values for association with MCL injury for each SNP. rs80351309 is located in the intergenic region between GRPEL1 and FLJ36777.



▶ **Fig. 4** Regional-association plot for rs6083471 with MCL injury. Tested SNPs are arranged by genomic position on chromosome 20 (x-axis) in a 300 kb window around the lead SNP rs6083471 (purple diamond). The y-axis indicates $-\log_{10}$ p-values for association with MCL injury for each SNP. rs6083471 is located 3' to the long non-coding RNA gene FLJ33581. The color of dots of the flanking SNPs indicates their linkage disequilibrium (R^2) with the lead SNP as indicated in the heat map color key. The red dot indicates rs79027201, which is tightly linked to rs6083471 ($R^2 > 0.99$).

Potential mechanisms underlying the association of rs80351309 and rs6083471 with MCL injury

Rs80351309 is located in the intergenic region between GRPEL1 (grpE protein homolog 1 mitochondrial precursor) and FLJ36777 (long non-coding RNA) on chromosome 4 (▶ **Fig. 3**). GRPEL1 is a homolog of GrpE protein, which is conserved from bacteria to humans and is involved in translocation of proteins into the mitochondria. FLJ36777 is a non-coding RNA with a length of 6 kb and one intron. rs6083471 is located 3' to FLJ33581, which is an uncharacterized long non-coding RNA of unknown function on chromosome 20 (▶ **Fig. 4**). There are at least 10 000 long non-coding RNAs in the genome. Several long non-coding RNAs have been shown to function in regulating gene expression, but the functions of FLJ36777 and FLJ33581 are unknown.

We searched for a mechanism whereby these SNPs or a linked SNP might affect the activity of nearby genes to account for their effects on MCL injury. For rs80351309, there are no other SNPs that are tightly linked using 1000 Genomes data that could serve as a causal mutation. rs6083471 and rs79027201 are tightly linked in the same linkage disequilibrium block on chromosome 20. Either one of these SNPs might be responsible for affecting risk for MCL injury, with the other one showing an association simply due to linkage.

However, neither rs80351309 nor either of the SNPs on chromosome 20 (rs6083471 or rs79027201) are in coding regions. Further, none of these SNPs has yet shown evidence of affecting expression of nearby genes as an expression quantitative trait locus in various cell lines or tissues [5].

Finally, we interrogated data from the ENCODE project regarding these 3 SNPs but found no evidence that any were in a DNAse I hypersensitive region or might affect binding of specific transcription factors [4].

Discussion

MCL injuries are one of the most common ligamentous injuries of the knee joint amongst athletes, often disrupting participation in sport. Preventative measures such as skill training and rule modifications have been proposed, however the efficacy of these meas-

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ures remains unknown [16, 17, 20]. Furthermore, biomechanical and clinical studies looking at the role of bracing in injury prevention show limited benefit [17–19].

While very little was previously known about the influence of genetic differences on MCL injury risk, the new information from this study holds promise in improving our fundamental understanding and prevention of this injury. By obtaining access to large-scale genotype and phenotype data from the RPGEH, we were able to find the first evidence for genetic factors affecting MCL injury. The data contained information from 102 503 individuals of whom 1 572 had an MCL injury. This genome-wide screen for SNPs associated with this injury provides new insights regarding the differences between individuals in their inherent propensity for injury.

Power calculations indicate that a cohort of this size would have about an 83 % chance of detecting a SNP with an association to MCL injury at genome-wide significance (assuming genotype relative risk of 1.7, minor allele frequency of 5 %). The chance of detection drops to 1 % for a SNP with a weaker effect on MCL injury (genotype relative risk of 1.2, minor allele frequency of 5 %).

MCL injuries are usually caused by an acute traumatic blow to the lateral aspect of the knee or a valgus moment without contact [24, 25]. While acute trauma is a leading risk factor, our data supports our hypothesis that genetic differences also appear to be a risk factor for MCL injury. Specifically, these genetic differences may play a role in whether a given blow to an individual results in an MCL injury, the grade of injury, as well as the rate of injury recovery.

Our cohort included people regardless of whether or not they participated in a sport. We cannot document whether the statistical association of rs80351309 and rs6083471 with MCL injury was derived predominantly from the subset of the population that were active in one or more sports. Furthermore, a portion of the injuries in our study may represent chronic, degenerative or overuse injury associated with the MCL [21, 27]. A less likely explanation for the association is that these polymorphisms are associated with an increased propensity to exercise and participate in sports that have a high risk of MCL injury.

SNPs rs80351309 on chromosome 4 and rs6083471 on chromosome 20 showed an association with MCL injury that was genome-wide significant. These results should be replicated in an independent population in future studies. For rs80351309, the association with MCL injury should be viewed with some caution because the genotypes were imputed with only moderate accuracy (overall $R^2 = 0.56$) and because it was the sole SNP in its region that showed an association with MCL injury.

Both SNPs are located in regions between genes with no previously known role in ligament structure and function. We have not yet found evidence that these sentinel SNPs or any SNPs that are closely linked within the same linkage disequilibrium block are directly responsible for affecting the function of nearby genes. None of these SNPs are in coding regions and none have yet been found to be expression quantitative trait loci that affect expression of nearby genes.

Individuals that have one copy of the risk allele for rs80351309 (T/C) or rs6083471 (C/T) have an increased risk of 2.1-fold or 1.6-fold compared to individuals lacking a risk allele, respectively. There are too few individuals homozygous for the risk allele for these SNPs to reliably measure risk in these groups. The frequency of the risk allele of these 2 SNPs is relatively low, so the risk for MCL injury for most people is not affected. However, for a specific individual harboring a risk allele, the effect of that SNP on risk for MCL injury is noteworthy.

In the future, it will be important to replicate the gene association results with MCL injury in an independent cohort. It will also be interesting to perform the analysis on a population of athletes competing in sports with high rates of MCL injury, such as American football, hockey, soccer, rugby or skiing. The results from these studies may reveal that certain genetic polymorphisms such as rs80351309 or rs6083471 could be used as diagnostic markers to predict which athletes harbor a higher risk for MCL injury. Preventative measures could then be taken to alleviate that risk, thereby reducing the overall incidence of injury.

There are several limitations to this study. First, the phenotypes were defined from codes contained in the electronic health records, which may be inaccurate. Second, the number of individuals of Latin-American, African-American and Asian ethnicity was relatively small, and hence the association results for these results are weaker than those from the European group. Third, additional studies are warranted to begin to illuminate the underlying biological mechanism for the association of variation near GRPEL1/FLJ36777 on chromosome 4 and FLJ33581 on chromosome 20 with MCL injury.

Conclusion

A genome-wide association screen for MCL injury revealed 2 SNPs (rs80351309 and rs6083471) with significant associations. Our data indicate that allelic variation at these 2 SNPs could explain part of the variation in risk for MCL injury. It will be important to replicate this finding in independent cohorts, including cohorts of athletes. In the future, sports medicine practitioners may be able to use this information to identify individuals at an increased risk, and subsequently take measures to prevent an MCL injury before it occurs.

What are the findings?

- We performed a genome-wide association study for MCL injury using data from Kaiser Permanente Northern California consisting of 1 572 cases and 100 931 controls.
- We discovered 2 SNPs (rs80351309 and rs6083471) associated with MCL injury at genome-wide significance ($p < 5 \times 10^{-8}$) with moderate effects (ORs = 2.12 and 1.57, respectively). Replication of this association in cohorts of athletes is warranted.

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Gladys Valley Foundation, and Kaiser Permanente. Development of genotypic data was supported by grant RC2 AG036607 from the National Institutes of Health. Information about data access can be obtained at: http://www.ncbi.nlm.nih.gov/projects/gap/cgibin/study.cgi?study_id=phs000674.v1.p1 and https://researchbank.kaiserpermanente.org/for-researchers/.

Conflict of Interest

The authors have no conflict of interest to declare.

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